claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a8130dd0f1780cfe1.jsonl`
- SHA-256: `384e020f89443e9fa5215bf5ac27938bfa649e9a603f4a35e5a496f7956948be`
- Source modified: `2026-05-31T04:21:05+00:00`
- Imported at: `2026-07-05T16:48:20+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:18:54.079Z)

Source Extractor

```
Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision /
document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud
API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date,
narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST
personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-
column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both
born-digital (text layer present) and the harder cases where pdfplumber's
extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies
between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default.
Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud
options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT,
BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-
commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This
is critical ? many document-AI models and their training datasets are non-commercial (e.g.
LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several
VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo
license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24
GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license
(state explicitly), latest release / maintenance status as of 2025-2026, VRAM need +
quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL
TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL
(note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 /
2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community
License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR,
EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND
PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),
MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON
```

transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Fetch and extract key claims from this source:
URL: https://arxiv.org/pdf/2511.14998
Title: FinCriticalED: A Visual Benchmark for Financial Fact-Level OCR (2025)
Found via: Borderless-table accuracy & financial-number hallucination/verification (contrarian/skeptical) search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-05-31T04:19:09.247Z)

I'll fetch the source and extract key claims. Let me start by loading the WebFetch tool schema.

3. assistant (2026-05-31T04:19:14.476Z)

I'll fetch the arxiv PDF and extract claims relevant to the research question about bank statement OCR and financial number hallucination/verification.

4. user (2026-05-31T04:19:28.299Z)

FinCriticalED: Financial Fact-Level OCR Benchmark - Key Findings

1. Benchmark Focus

The benchmark evaluates optical character recognition and document parsing on **financial documents**, specifically assessing accuracy at the fact (cell/value) level rather than character level. It targets real-world financial tables with challenges like dense formatting and complex layouts.

2. Systems Evaluated

Vision-Language Models (VLMs):
- Qwen models (Qwen3, Qwen2.5, Qwen2-VL)
- Claude (Anthropic)
- GPT-4o (OpenAI)
- Gemini (Google)
- DeepSeek models
- InternVL
- LLaMA-based models
- MiniCPM

Traditional OCR & Parsing Tools:
- PaddleOCR
- Tesseract
- docTR
- Marker
- GOT-OCR
- Cloud services: AWS Textract

3. Key Quantitative Findings

The document references performance comparisons across models but the compressed PDF stream limits exact metric extraction. References indicate "fact-level accuracy" as the primary metric, with models showing varying performance on financial data extraction tasks.

4. Error Types in Financial OCR

The paper identifies critical issues in financial document processing:
- **Hallucinations** in generated content
- **Transposed digits** in numerical values
- **Dropped or duplicated rows** in tables
- **Numerical errors** affecting fact accuracy

5. Challenges in Financial Documents

Financial documents present unique difficulties:
- Dense table structures
- Multi-column layouts
- Borderless or semi-ruled formats
- Complex cell relationships requiring precise alignment

6. VLM vs. Traditional OCR Comparison

The benchmark compares modern vision-language models against traditional OCR pipelines, evaluating which approach better handles financial document complexity.

7. Proposed Metric

The paper proposes **fact-level error detection/accuracy** as the evaluation metric, moving beyond character-error-rate to assess whether critical financial facts are correctly extracted.

Publication: arXiv:2511.14998v3 (April 2026)
Authors: Led by Yueru He, Sophia Ananiadou and team

[Binary content (application/pdf, 2.2MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\tool-results\webfetch-1780201161953-fwgfxf.pdf]

5. assistant (2026-05-31T04:19:31.375Z)

The WebFetch gave a useful overview but the compressed PDF stream limited exact metric extraction. The PDF was saved locally. Let me read the actual PDF to extract verbatim quotes and specific numbers.

6. user (2026-05-31T04:19:32.432Z)

```
pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)
```

7. assistant (2026-05-31T04:19:34.836Z)

```
PDF rendering needs poppler. Let me check if I can extract text another way using Python (pypdf
is already a dependency of this project per the git status).
```

8. user (2026-05-31T04:19:36.270Z)

```
Exit code 1
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import pypdf; print(pypdf.__version__)
    ^^^^^^^^^^^^
ModuleNotFoundError: No module named 'pypdf'
---
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import pdfplumber; print('pdfplumber', pdfplumber.__version__)
    ^^^^^^^^^^^^^^^^^
ModuleNotFoundError: No module named 'pdfplumber'
```

9. assistant (2026-05-31T04:19:38.641Z)

```
Not in the base environment. This project uses a virtualenv likely. Let me check for a venv and
try extracting text from the PDF.
```

10. user (2026-05-31T04:19:41.177Z)

```
C:/Users/avidu/Projects/muneem/.venv/Scripts/python.exe
===
ok 6.12.2 0.11.9
```

11. assistant (2026-05-31T04:19:43.007Z)

```
The project venv has pdfplumber and pypdf. Let me extract the text from the saved PDF.
```

12. user (2026-05-31T04:19:45.897Z)

```
NUM PAGES: 13
FinCriticalED: A Visual Benchmark for Financial Fact-Level OCR
YueruHe YupengCao YanWang
ColumbiaUniversity StevensInstituteofTechnology TheFinAI
USA USA USA
yh3507@columbia.edu
LingfeiQian ShuyaoWang YiHan
TheFinAI TheFinAI GeorgiaInstituteofTechnology
USA USA USA
HaohangLi RuoyuXiang FanZhang
StevensInstituteofTechnology NewYorkUniversity TheUniversityofTokyo&MBZUAI
USA USA Japan&UAE
ZhuohanXie MingquanLin PrayagTiwari
MBZUAI UniversityofMinnesota HalmstadUniversity
UAE USA USA
XueqingPeng? JiminHuang GuojunXiong
TheFinAI TheFinAI HarvardUniversity
USA USA USA
xueqing.peng2024@gmail.com
SophiaAnaniadou
UniversityofManchester
UK
Figure1:OverviewofFinCriticalED.Left:standardOCRmetricsfailtocapturefinanciallycriticalsemantice
rrors.Middle-left:
constructionofafact-centricdatasetwithfivetypesofannotatedfinancialfacts.Middle-
```

right:evaluationpipelineusing
OCR/VLM/MLLMoutputswithDeterministic-Rule-GuidedLLM-as-
Judge.Right:Keyinsightsfromexperimentson13OCR
systemsandMLLMs.
Abstract
whetherOCRandvision-languagesystemspreservefinancially
Recentprogressinmultimodallargelanguagemodels(MLLMs)has critical evidence beyond lexical
similarity. FinCriticalED con-
substantiallyimproveddocumentunderstanding,yetstrongoptical tains859real-
worldfinancialdocumentpageswith9,481expert-
characterrecognition(OCR)performanceonsurfacemetricsdoes
annotatedfactsspanningfivecriticalfieldtypes:numeric,temporal,
notguaranteefaithfulpreservationofdecision-criticalevidence.
monetaryunit,reportingentity,andfinancialconcept.Weformu-
Thislimitationisespeciallyconsequentialinfinancialdocuments, latethetaskasstructuredOCRwithfact-
levelverification,and
wheresmallvisualerrorscaninducediscreteshiftsinmeaning.To developaDeterministic-Rule-GuidedLLM-
as-Judgeprotocoltoas-
studythisgap,weintroduceFinCriticalED(FinancialCritical
sesswhethermodeloutputspreserveannotatedfactsincontext.We
ErrorDetection),afact-centricvisualbenchmarkforevaluating
benchmark13systemsspanningOCRpipelines,specializedOCR
VLMs,open-sourceMLLMs,andproprietaryMLLMs.Resultsreveal
?Correspondingauthor.
6202
rpA
7
]VC.sc[
3v89941.1152:viXra

13. assistant (2026-05-31T04:19:49.271Z)

I have the abstract. The text extraction is working (though words run together due to font encoding). Let me get the key results pages ? the abstract continues, plus the experiments/results tables. Let me grab the rest of page 1 and search for the quantitative findings.

14. user (2026-05-31T04:19:51.720Z)

(Bash completed with no output)

15. user (2026-05-31T04:19:56.236Z)

LEN P1: 2097
''

16. assistant (2026-05-31T04:19:58.951Z)

Page 1 was only 2097 chars (the abstract was cut at "benchmark13systems..."). The rest is on subsequent pages. Let me pull pages 2 and the results section. Let me search across all pages for the key result terms and dump the most relevant pages.

17. user (2026-05-31T04:20:05.711Z)

```
0 len 2097 hits: []
1 len 5666 hits: ['marker', 'f1', 'accuracy']
2 len 5708 hits: ['gemini', 'claude', 'paddleocr', 'textract', 'deepseek']
3 len 5709 hits: ['gemini', 'gpt-4o', 'claude', 'qwen', 'textract', 'accuracy', 'deepseek']
4 len 5613 hits: ['gemini', 'gpt-4o', 'claude', 'qwen', 'marker', 'accuracy', 'best',
'outperform', 'deepseek']
5 len 5209 hits: ['gpt-4o', 'marker', 'hallucinat', 'deepseek']
6 len 9342 hits: ['gemini', 'gpt-4o', 'claude', 'qwen', 'paddleocr', 'deepseek']
7 len 5991 hits: ['marker']
8 len 4640 hits: ['accuracy']
```

```
9 len 6417 hits: ['gpt-4o', 'accuracy']
10 len 5256 hits: ['gpt-4o', 'paddleocr', 'accuracy', 'hallucinat']
11 len 4196 hits: ['gpt-4o', 'claude', 'qwen']
12 len 4093 hits: ['gpt-4o', 'accuracy', 'hallucinat', 'deepseek']
```

18. assistant (2026-05-31T04:20:08.496Z)

```
The richest results pages are 4 (best/outperform), 5 and 10/12 (hallucination), 3 and 6 (model
lists/tables). Let me extract these key pages.
```

19. user (2026-05-31T04:20:11.327Z)

```
========== PAGE 4 ==========
FinCriticalED:AVisualBenchmarkforFinancialFact-LevelOCR ?,,
Table5:ModelperformancesfortraditionalOCRpipelines,
specializedOCRVLMs,open-sourceMLLMs,andproprietary
MLLMs.Boldindicatesthebestperformanceineachcolumn,
whileunderlinedvaluesdenotethesecond-best.Forerror
rate(E?),lowerisbetter.Fact-levelmetricsincludenumerical
(N-FFA),temporal(T-FFA),monetaryunit(MU-FFA),report-
ingentity(RE-FFA),andfinancialconcept(FC-FFA)accuracy;
FFAdenotesoverallfinancialfactaccuracy.
Model Size General(%) Fact-Level(%)
ROUGE-1? ROUGE-L? EditDistance? N-FFA? T-FFA? MU-FFA? RE-FFA? FC-FFA? FFA?
OCRPipelines
MinerU2.5[23] 1.2B 95.71 95.30 6.02 98.76 96.48 54.05 91.09 96.44 94.64
PP-OCRv5[4] 0.07B 97.54 96.55 3.10 95.70 90.29 90.00 86.62 93.75 91.91
SpecializedOCRVLMs
DeepSeek-OCR[35] 6B 94.73 94.42 7.33 93.47 91.96 83.53 92.27 94.36 92.67
DeepSeek-OCR-2[36] 3B 92.90 92.18 10.72 82.63 91.90 82.83 88.69 86.51 86.19
GLM-OCR[5] 0.9B 95.10 94.74 6.43 93.24 98.53 88.89 97.84 100.00 96.92
Open-sourceMLLMs
Gemma-3n-E4B-it[10] 4B 83.49 79.59 23.82 52.65 77.06 64.71 74.65 72.86 65.68
Qwen3-VL-8B-Instruct[29] 8B 97.68 97.40 2.93 98.47 96.99 97.65 93.18 99.24 96.88
Llama-4-Maverick[21] 17B 98.00 97.62 3.70 97.77 97.99 97.65 94.26 98.48 96.92
Qwen3.5-397B-A17B[28] 397B 98.12 98.00 2.59 87.72 87.99 86.14 91.22 94.40 89.70
ProprietaryMLLMs
GPT-4o[12] - 90.40 88.35 16.01 59.56 84.59 81.92 81.78 70.84 71.68
GPT-5[24] - 91.81 89.56 15.79 66.83 94.48 92.35 89.19 91.77 81.65
Claude-Sonnet-4.6[1] - 98.84 98.73 1.69 98.59 97.99 97.06 94.02 98.94 97.23
Figure2:ModelperformanceongeneralOCRmetricsand
Gemini-2.5-Pro[9] - 98.81 98.37 2.46 97.24 97.82 97.65 94.18 98.94 96.74
financial fact-level accuracies. Best results per metric are
highlightedwithastar(?).
4.1 MainResultsandDiscussion
4.1.1 GeneralOCRMetricsvs.FinancialFidelity. RQ1:Towhat
importantadvantageforfinanciallyfaithfultranscription,although
extentdoconventionalOCRmetricsreflectfinanciallyfaithfulex-
thestrongestgeneralMLLMscannowmatchorevenexceedit.
tractionofcriticalfields?
4.1.3 DifficultyAcrossFinancialCriticalFactTypes. RQ3:Which
TheresultsshowthatconventionalOCRmetricsareinsufficient
typesoffinancialcriticalfactsaremostvulnerabletoOCRerrors
forevaluatingfinancialfidelity.Whiletheycaptureoverallrecon-
acrosscurrentmodels?
structionquality,theydonotreliablyreflectwhetherfinanciallycrit-
Theresultsshowthatnumericalvaluesandmonetaryunitsare
icalfieldsarepreservedfaithfully.Forexample,MinerU2.5achieves
themostvulnerablefacttypesacrosscurrentmodels,whiletem-
anear-ceilingROUGE-1scoreof95.71%andROUGE-Lscoreof
poral facts and financial concepts perform relatively better but
95.30%,yetdropssharplyto54.05%onMU-FFA,indicatingthat
remainfarfromrobust.Forexample,Gemma-3n-E4B-itachieves
stronglexicaloverlapcanstillmaskseriousfact-levelerrors.Similar
only52.65%,GPT-4o59.56%,andGPT-566.83%onnumericalfacts,
```

gapsremainevenamongthestrongestmodels:Claude-Sonnet-4.6
indicatingasubstantialgapinpreservingreliablenumericaccu-
reaches98.84%ROUGE-1and98.73%ROUGE-Lbutonly94.02%
racy?anessentialrequirementforfinancialdocumentunderstand-
onRE-FFA,whileQwen3.5-397B-A17Battains98.12%ROUGE-1
ing.MinerU2.5reachesonly54.05%onMU-FFA,suggestingthat
and98.00%ROUGE-Lyetfallsto86.14%onMU-FFA.Theseresults
nearlyhalfofmonetaryunitfieldsareincorrectlyrecognized,which
showthathighoverlap-basedOCRscoresdonotnecessarilyimply
canseverelydistortfinancialmeaning.EvenClaude-Sonnet-4.6,
financiallyfaithfulextraction,especiallywhencorrectnessdepends
despiteitsmorebalancedoverallperformance,stillfallsshortof
onexactvalues,units,reportingsubjects,andfieldassociations.
thenear-perfectaccuracyrequiredforreal-worldfinancialappli-
4.1.2 SpecializedOCRModelsvs.GeneralMLLMs. RQ2:Howdo
cations.Notably,eventhecomparativelystrongerfactcategories
specializedOCRmodelsandgeneral-purposeMLLMsdifferintheir
failtoreachtrustworthylevels.Forinstance,Qwen3.5-397B-A17B
abilitytopreservefinanciallycriticalinformation? achieves87.99%onT-FFAand94.40%onFC-
FFA,whileweaker
Inourbenchmark,OCR-specializedmodelsshowstrongperfor-
modelssuchasGemma-3n-E4B-itdropto52.65%onN-FFAand
mancebutnotuniformlydominantonfinanciallycriticalextraction. 77.06%onT-
FFA.Overall,theseresultssuggestthatcompactlocal
AmongtheOCR-specializedsystems,GLM-OCRachieves96.92%
cues,suchasdigits,decimalpoints,signs,andunitmarkers,are
FFA,outperformingPP-OCRv5at91.91%andDeepSeek-OCR-2at themosterror-
sensitiveelementsinfinancialOCR,andthateven
86.19%,whichindicatesthatOCR-orientedmodelsremaincom-
relativelystrongerfacttypeshavenotyetachievedthelevelof
petitivebaselinesforexactfieldpreservation.Atthesametime, reliabilityrequiredforhigh-
stakesfinancialinterpretation.
severalgeneralMLLMsachievecomparableorstrongerfact-level
4.2 ModelFailureAnalysis
performance,includingQwen3-VL-8B-Instructat96.88%,Llama-4-
Maverickat96.92%,Gemini-2.5-Proat96.74%.Amongthesemodels, Beyondaggregatefact-
levelaccuracy,weconductapage-leveler-
Claude-Sonnet-4.6getsthebestresultwith97.23%.Itshowsthe roranalysisof
306OCRcasesfrom6representativemodels,all
greatpotentialofMLLMsinOCRtasks.However,thiscapabilityis
reviewedbyfinancialexperts.WhileFFAevaluateswhetheran-
notconsistentacrossthebroadergeneral-MLLMgroup:Gemma-3n-
notatedfinancialcriticalfactsarepreserved,expertreviewfurther
E4B-itreachesonly65.68%FFA,GPT-4o71.68%,andGPT-581.65%, capturesnon-
criticalOCRerrors,layoutdistortions,andotherpage-
all below the strongest OCR-oriented and multimodal systems.
levelfailuresthatstillaffectpracticaluse.Thismeansthatamodel
Overall,thesefindingssuggestthatOCRspecializationremainsan
mayachievehighFFAorROUGEscoresyetstillexhibitsubstantial
========== PAGE 5 ==========
?,, Heetal.
Table 6: Page-level error statistics across models. Critical
error: if at least one annotated financially critical field is
incorrect;non-criticalerror:ifOCRdiscrepanciesoccurout-
sideannotatedcriticalspans.

Model	CriticalError Rate(%)?	Non-CriticalError Rate(%)?	CleanPages (%)?	CriticalOnly (%)?	Non-Critical Only(%)?	Both (%)?
OCRPipeline						
MinerU2.5[23]	23.5	58.8	32.4	8.8	44.1	14.7
SpecializedOCRVLMs						
DeepSeek-OCR[35]	17.6	86.3	13.7	0.0	68.6	17.7
Open-sourceMLLMs						
Gemma-3n-E4B-it[10]	74.5	90.2	7.8	2.0	17.7	72.6
Llama-4-Maverick[21]	15.7	49.0	49.0	2.0	35.3	13.7
ProprietaryMLLMs						

GPT-4o[12] 52.9 80.4 11.8 7.8 35.3 45.1
GPT-5[24] 58.0 86.0 12.0 2.0 30.0 56.0
page-levelerrors.Weanalyzethesefailuresalongthreeaxes:(i)
financialseverity,distinguishingbetweencritical-fieldandnon- Figure3:Page-
levelcriticalerrorrate(%)bydocumentcom-
criticalOCRerrors;(ii)documentmodalityandcomplexity,
plexityandmodality.Eachsubtablereportsa3×3breakdown
examininghowperformancevariesacrosspagetypesanddifficulty
whererowsdenotecomplexity(Low,L;Medium,M;High,H)
levels;and(iii)model-specificfailurepatterns,characterizing andcolumnsdenotemodality(text-
only,T;table-only,Tb;
recurringerrorbehaviorsacrossmodelfamilies. mixture,M).Greycellwith"-"represents0sampleunderthis
category.
Financial severity. We define a critical error as any OCR
mismatchinvolvingoneofthefiveannotatedcriticalfieldtypes,
andanon-criticalerrorasanyOCRdiscrepancyoutsidethese
punctuationordashes.Thisisconsistentwiththeirarchitectures:
fields,suchaserrorsinsurroundingnarrativetext,punctuation, MinerU2.5usesacoarse-to-finetwo-
stagepipelinethatseparates
orspacing.Table6showslargedifferencesinpage-levelfinancial
layoutanalysisfromlocalrecognition[23],whileDeepSeek-OCR
reliability:thestrongestsystemsmakecriticalerrorsononly17.6%?
reliesonopticalcompressiontorepresenthigh-resolutionpages
23.5%ofpages,whileweakerbaselinesdosoonnearlythree-fourths
withcompactvisualtokens[35].Bycontrast,weakerormoregen-
ofthebenchmark.Non-criticalOCRerrorsremaincommonacross
erativemultimodalLLMsproducemorediverseandlesspredictable
almostallmodels,rangingfrom49.0%to90.2%,showingthatstrong
errors.InGemma-3n-E4B-itandGPT-4o,weobservehallucinated
factpreservationdoesnotimplycleantranscription.
continuation,wherethemodelincorrectlycompletesasentence
Adecompositionintoclean,non-criticalonly,criticalonly,and
usingpreviouslyseentextwithasimilarprefix,alongwithcontext-
bothfurtherhighlightsthisgap.Llama-4-Maverickyieldsthehigh- driven substitutions and unstable
free-form corruption [10, 12].
estshareofcleanpages(49.0%),whereasGemma-3n-E4B-ithasthe
StronggeneralmodelssuchasLlama-4-Maverickremainmoresta-
largestshareofpagescontainingbothcriticalandnon-criticaler-
bleatthepagelevel,suggestingadistinctionbetweengrounded
rors(72.6%).Thus,practicalfinancialOCRrequiresminimizingboth
transcriptionandsemanticreconstruction.
materiallydangerouserrorsandbroadertranscriptionartifacts.
5 Conclusion
Documentmodalityandcomplexity. Wenextstratifypage-
Inthiswork,weintroducedFinCriticalED,afact-centricvisual
levelcriticalerrorratesbydocumentmodalityandvisualcomplex-
benchmarkforevaluatingwhetherOCRsystemsandMLLMspre-
ity.Pagesaregroupedintotext-only,table-only,andmixedmodali-
servefinanciallycriticalevidenceinreal-worlddocuments.The
ties,andintoLow,Medium,andHighcomplexitylevels(definitions
benchmarkincludesanexpert-annotateddatasetoffinanciallycrit-
areintheAppendix).
icalfieldsandanLLM-poweredevaluationprotocolforstructured
AsshowninFigure3,mixedpagesareconsistentlythehardest
fact-levelassessment.Ourexperimentsyieldthreemainfindings:
setting.Weakergeneral-purposeMLLMsexhibitveryhighcriti-
(1) Conventional text similarity metrics do not reliably capture
calerrorratesinthiscategory,withGemma-3n-E4B-itreaching
financialfaithfulness:highsurface-leveloverlapcanstillmasksub-
100.0% across all complexity levels and GPT models remaining
stantialfact-levelerrors.(2)Notallcriticalfieldsareequallydifficult:
at80.0%?100.0%.Bycontrast,OCR-specializedmodelsaremore
numericalfieldsandmonetaryunitsareconsistentlymorefrag-
stable,thoughnoterror-free:ontext-onlypages,DeepSeek-OCR
ilethantemporal,reporting,andconceptfields,showingthatthe

risesfrom0.0%atlowcomplexityto23.1%athighcomplexity,and
mainchallengeliesinpreservingsmalllocalcuessuchasdigits,
MinerU2.5risesfrom0.0%to38.5%.Table-onlypagesaregenerally
signs,andunitmarkers.(3)OCR-specializedsystemsremainstrong
easierwhenlayoutsaresimple.Overall,themainchallengeisOCR
and stable baselines, but the strongest MLLMs can now match
underheterogeneousstructureratherthanOCRalone.
orexceedthem,suggestingthatgroundedtranscriptionbehavior
Model-specificfailurepatterns. Beyondaggregateerrorrates,
mattersmorethanmodelfamilyalone.Overall,FinCriticalED
modelfamiliesexhibitdistinctfailuremodes.OCR-specializedsys- reframesfinancialOCRasafact-
preservationproblemratherthan
temssuchasDeepSeek-OCRandMinerU2.5mostlymakefixed,
asurfacetranscriptiontask.Wehopethebenchmarkwillsupport
repetitiverecognitionerrors,includingrecurringlexicalcorruptions
futureresearchonfinanciallyfaithful,layout-awaredocumentun-
(e.g.,transferor?transfessor)andstableformattingartifactsaround derstandinginprecision-
criticaldomains.

20. assistant (2026-05-31T04:20:14.734Z)

This is excellent ? I now have the full results table and the key narrative findings. This is a rich, directly-relevant source. Let me grab the model lists/setup (page 3) and the hallucination discussion (pages 10, 12) to confirm specifics and the LLM-as-Judge / verification angle.

21. user (2026-05-31T04:20:20.278Z)

========== PAGE 3 ==========
?,, Heetal.
Table4:Statisticsandfinancialcriticalfactsdistributionof
and(3)anyentitypassinganormalizedexact-matchcheckisdeter-
theFinCriticalEDdataset. ministicallyforcedtocorrect=true,guardingagainstLLMfalse
Type Count AverageperDocument Percentage(%)
negatives.Ratherthanpenalizingsuperficialformattingdifferences,
DatasetSize 859 ? ? thejudgefocusesonsemanticfidelity:itdetermineswhetherthe
NumericFacts(??) 2,930 3.4 30.9 modelhassuccessfullylocatedandreproducedeachentity,and
TemporalFacts(??) 1,144 1.3 12.1 whetherthereproducedvalueisfactuallyexact.Conditionedona
MonetaryUnits(???) 881 1.0 9.3 structuredevaluationprompt(seeinAppendix),theDeterministic-
ReportingEntities(???) 3,142 3.7 33.1
FinancialConcepts(? ??) 1,384 1.6 14.6 Rule-GuidedLLM-as-Judgeoutputsper-
fieldcorrectnessdecisions
Total(?) 9,481 11.0 100.0 usedtocomputeaccuracymetricsacrosscriticalfieldstypes.
Torigorouslyvalidatetheautomatedevaluator,weconducteda
human?LLMalignmentvalidationbysampling15modeloutputs
3.2 FinancialFactEvaluation
eachfromPP-OCRv5,GPT-4o,andGPT-5.Weadoptastringent
TraditionalOCRevaluationreliesonsurface-levellexicalmetrics Document-
LevelStrictAgreementmetric:theLLMjudgeandhuman
(e.g.,ROUGE,editdistance)thatrewardsuperficialstringoverlap.
expertareconsideredinagreementifandonlyiftheircorrectness
However,thesemetricsfailtoassesswhetheramodelpreservesthe decisions(?
?)matchperfectlyacrossallentitieswithinadocument.
financialmeaningofadocument.Infinance,evensingle-character Evenasingleentity-
leveldivergenceresultsinadocument-level
OCRdeviationscanaltertheunderlyingreality.FinCriticalED
disagreement.Followingiterativepromptandworkflowrefinement,
operationalizesashiftfromtextsimilaritytofact-levelcorrectness,
theevaluationprotocolachievesarobust95%agreementrate,con-
measuringtheexactpreservationofdomain-criticalfactualcontent. firmingitsreliabilityasadomain-
awarejudgeforcomplexfinancial
factualverification.
3.2.1 FinancialFactAccuracy(FFA)Formulation. Theobjectiveis
toverifywhetheramodelcorrectlyreproducestheannotatedfacts
3.3 ExperimentalSetup
? =? ??? ??? ???? ???? ?? fromtheground-truthHTMLwithin

themodel-generatedoutput.Foreachfact? ? ??,weassignabinary 3.3.1 EvaluationModels.
Weevaluatefourcategoriesofsystemsto
correctnessindicator? ? ? {0,1},where? ? = 1if ? ? isrecovered
comprehensivelyassessfinanciallycriticaldocumentunderstand-
inasemanticallyequivalentlocalcontext,and0otherwise.Cru- ing:
cially,factualmatchingacrossallcategoriesrequirestheabsolute
OCRPipelines.Modularsystemsthatdecoupletextdetection,
preservation of visually subtle but determinant cues, including
recognition,anddocumentreconstruction,includingMinerU2.5[23]
signs,decimals,unitscales,temporalboundaries,andentityattri- andPP-
OCRv5[4].Theseserveasstrongengineering-orientedbase-
bution.Forexample,converting?(1,200)?to?1,200?yields? ? =0,as linesforstructureddocumentparsing.
theomissionoftheparentheticalnegativenotationfundamentally
SpecializedOCRVLMs.Specializedfoundationmodelsoptimized
invertsthefinancialmeaning. forvisuallygroundedtextextractionandstructure-awarerecog-
For a given output document, the overall Financial Fact Ac- nition,includingDeepSeek-OCR
[35],DeepSeek-OCR-2 [36],and
curacy(FFA)iscomputedas? = (cid:205) ? ? ?/|?|.Wesimilarlydefine GLM-
OCR[5].TheseprovideatightlyintegratedOCR-centrical-
category-specificaccuraciesas? ? =(cid:205) ????? ? ?/|? ?|foreachsubset
ternativetotraditionalpipelines.
? ? {?,?,??,??,??}.Toobtainrobustdataset-levelperformance Open-
SourceMLLMs.Unifiedmultimodalmodelsthatjointly
normalizedacrossdocumentswithvaryingfactdensities,weag-
performvisualunderstandingandlanguagereasoning,including
gregatethetotalcorrectcounts? ? andtotalfactcounts? ? across
Gemma-3n-E4B-it[10],Qwen3-VL-8B-Instruct[29],Llama-4-Mav-
allsamples.Theglobalbenchmarkmetricforeachcategoryisthus erick[21],andQwen3.5-397B-A17B[28].
definedasx-FFA=? ?/? ?,yieldingdistinctevaluationsfornumeric ProprietaryMLLMs.Frontierclosed-
sourcemodelswithstrong
(N-FFA),temporal(T-FFA),monetaryunit(MU-FFA),reporting multimodal reasoning capabilities,
including GPT-4o [12], GPT-
entity(RE-FFA),andfinancialconcept(FC-FFA)preservation.The 5[24],Claude-
Sonnet-4.6[1],andGemini-2.5-Pro[9].Theseserveas
overalldataset-levelFFAiscomputedanalogouslyoverallfactsin upper-boundreferencesforend-to-
enddocumentunderstanding
?,providingaunifiedmeasureoffactualfidelity. withoutexternalOCR.
3.2.2 Deterministic-Rule-GuidedLLM-as-JudgeVerification. Because
3.3.2 ImplementationDetails. Allproprietarymodelsareaccessed
financialfactsdonotmapcleanlyontosurfaceforms,traditional
viatheirofficialAPIswithtemperaturesettozerofordetermin-
stringoverlapisfundamentallyinsufficientforevaluation.Asingle
isticoutputs.Open-sourceMLLMsarehostedbyTogetherAIAPI.
factmayadmitmultiplevalidexpressions(e.g.,?$1.2B?vs.?1,200
OCRpipelinesandspecializedOCRVLMsareeitherself-hosted
million?),whereasalexicallyminordeviationcanentirelyfalsify
onfourNVIDIAGPUsoraccessedviatheirpublicAPIendpoints.
thepreservedinformation(e.g.,?16,950?vs.?16.950?).Toevaluate
Allmodelsshareaunifiedinput?outputformatalignedwiththe
atthefactlevel,weemployaDeterministic-Rule-GuidedLLM-as-
FinCriticalEDannotationschema.
Judgepipelinecombiningdeterministicparsingwith LLM-powered
semanticjudgmentacrossthefivefacttypes.Thepipelineproceeds
4 ResultsandAnalysis
inthreestages:(1)Extractallannotatedentitiesfromground-truth
HTMLwith±40-charactercontexthintsfordisambiguation;(2)
ToassesswhethercurrentOCRandMLLMsystemstrulypreserve
LLM?as-Judgeframework(poweredbyGPT-4o)determinesper
financiallycritical,visuallygroundedevidence,weorganizeour
entitywhetheritwasfoundandexactlymatchedintheprediction;
experimentsaroundthefollowingthreeresearchquestions.
========== PAGE 10 ==========

judgeandhumanmustagreeoneveryentitywithinadocumentto
countasaligned;asingledivergenceconstitutesdisagreement.GPT-

*total_entitiescountsMUSTmatchthecountsfromthepre-extractedlist.
*REMINDER:ifmatched_textequalsvalue(normalized),correctMUSTbetrue?
4oachieved95%agreementafteriterativepromptandevaluation
donotsetcorrect=falsejustbecausethesurroundingcontextinthe
predictiondiffersfromcontext_hint. workflowrefinement.Figure5showsarepresentativealignment
""" example:
Case:OverallFFA=90%. OnFigure5,bothhumanexpertsandthe
LLM-as-Judgeoutputsample
LLM-as-Judgeclassifytheoutputasfactuallyunreliable.TheLLM-
as-Judgehallucinatestherowname"Certificaterate"to"Interest
{'total_entities':65, rate,"asitassociatewiththe"Interestpaid"rownamebelow.The
'total_entities_with_Number_type':37,
'total_entities_with_Temporal_type':10, modelalsomisreadthemonetaryamount.LLMoutput:
'total_entities_with_Monetary_Unit_type':7,
'total_entities_with_Reporting_Entity_type':11,
'total_entities_with_Financial_Concepts_type':0, {
'found_entities':64, "total_entities":23,
'found_entities_with_Number_type':36, ...
'found_entities_with_Temporal_type':10, "found_entities_with_Number_type":3,
'found_entities_with_Monetary_Unit_type':7, ...
'found_entities_with_Reporting_Entity_type':11,
"found_entities_with_Financial_Concepts_type":10,
'found_entities_with_Financial_Concepts_type':0, ...
'correct_entities':64, "correct_entities_with_Number_type":2,
'correct_entities_with_Number_type':36, ...
'correct_entities_with_Temporal_type':10, "correct_entities_with_Financial_Concepts_type":9,
'correct_entities_with_Monetary_Unit_type':7, "entity_accuracy":90.0,
'correct_entities_with_Reporting_Entity_type':11,
"overall_explanation":"Themodelaccuratelyidentifiedmostentities,
'correct_entities_with_Financial_Concepts_type':0,
especiallyreportingentitiesandfinancialconcepts,butstruggled
'entity_accuracy':98.36, withexactnumericmatches."
'overall_explanation':'Themodelsuccessfullyidentifiedandcorrectly }
matchednearlyallentities,withonlyoneminordiscrepancyinanumber
format.',
'entity_diagnostics':[{'type':'Number', Humanexpertexplanation:
'value':'10%',
'context_hint':'XDirector10%OwnerOfficer(givetitlebelow)Other',
'found_in_prediction':True, Totalentities:23
'correct':True, Numericentities:3
... ...
'not_found_entities_with_Number_type':1, FinancialConcepts:10
'not_found_entities_with_Temporal_type':0, ...
'not_found_entities_with_Monetary_Unit_type':0, Wrongnumericentities:1
'not_found_entities_with_Reporting_Entity_type':0, ...
'not_found_entities_with_Financial_Concepts_type':0} Wrongreportingentities:1
Overall: Themodelperformswelloverall,excepterrorsduetohalluciantion
on"Interestrate"andwrongnumericalentityof"21,829,000
(shouldbe"21,429,000").
DesignDetails. ContextWindow.Manyfinancialdocumentsre-
peatidenticalvaluesacrossdifferentlineitems(e.g.,thenumber
Theseobservationscorrespondpreciselytotheerrorsandoverall
?1,200?mayappearinbothrevenueandheadcountrowsofthe judgmentfromGPT-4ooutput.
sametable).Topreventthejudgefromconflatingthese,eachentity
isaccompaniedbythe40charactersimmediatelyprecedingand D ModelFailureAnalysis
followingitinthegoldtext.Forinstance,theentity1,200mightbe
Thissectionanalyzesmodelfailuresbeyondoverall FFAscores.We
disambiguatedbythehint?...total revenue (in millions)
firstdescribetheerror-analysissetup,thenexaminetwonotable
1,200 for the fiscal year...?,directingthejudgetoverify
aggregatepatterns:elevatedmissing-outputbehaviorinsomepro-
thatspecificoccurrenceratherthananysurfacematch.
prietarymodelsandGLM-OCR?sstrongperformanceonfinancial
NormalizationandDeterministicOverride.Beforeapplyingthe
concepts.Wefinallypresentrepresentativeerrorcasesillustrating

override, both the predicted and gold entity values are normal-
howdifferentfailuremodesappearinpractice.
ized:lowercased,whitespace-collapsed,andstrippedofpunctua-
tion(commas,currencysymbols,trailingperiods).Ifthenormalized D.1 ErrorAnalysisSetup
stringsareidentical,theentityisforcedcorrect.Thisguardsagainst
Weperformerroranalysisbycomparingannotatedground-truth
LLMfalsenegativesonunambiguouscases.
contentwithmodel-predictedOCRoutputsafterbotharerendered
Foundvs.Correct.Anentityisfoundifthejudgelocatesit(ora
inHTML.Specifically,theground-truthdocumentandthecorre-
semanticallyequivalentexpression)anywhereintheprediction;itis
spondingmodeloutputareloadedintoLabelStudio,whereannota-
correctifthereproducedvalueisadditionallyjudgedtobefactually
torsinspectthemsidebysideandidentifyfinanciallymeaningful
exact.Anentitymaybefoundbutincorrect(e.g.,anumberlocated
discrepancies.Duringreview,theyassesswhetheramodelpre-
inthewrongcontext),orneitherfoundnorcorrect(entirelyabsent
servesbothlocaltranscriptionandthefinancialmeaningofeach
fromtheprediction).
criticalfieldunderitsrenderedcontext,ratherthanonlysurface-
HumanAlignmentValidation. Tovalidatetheautomatedjudge, leveltextualoverlap.
15 outputs from Paddleocrv5, gpt-4o, and gpt-5 each (45 total)
Weorganizereviewedexamplesalongtwodimensions:docu-
wereindependentlyevaluatedbyahumanexpertusingthesame
mentmodalityanddocumentcomplexity.Wefirstcategorize
found/correctschema.WeadoptDocument-LevelStrictAgreement: documentsintothreemodalitytypes:
========== PAGE 12 ==========
FinCriticalED:AVisualBenchmarkforFinancialFact-LevelOCR ?,,
Table8:QualitativetaxonomyofobservedModelOCRfailurepatterns.OCR-
specializedsystemstendtoshowfixedand
repetitiverecognitionerrors,whilegeneralmultimodalLLMsexhibitmorevariableandgenerativefailuremod
es.
ErrorType Description CommonModels Example
Header misalign- Correctvalueistranscribedbutattached Llama-family
"6.OwnershipForm:Direct(D)orIndirect(I)(Instr.
ment tothewrongcolumnorlocalheadercon- 4)"valuealignedto"7.NatureofIndirectBeneficial
text. Ownership(Instr.4)".
Headingrecogni- Themodelfailstocorrectlyrecognize Allwithdifferentlevelsof
gemma-3n-E4B-itfailedtorecognizesectionsubtitle
tionerror sectiontitlesorsubtitles,eitherbyomit- severity
"Creditcardratesmaydeclinewithoutacorrespond-
tingthemortranscribingthem incor- ingchangeintheamountsneededtopaythecertifi-
rectly, leading to incorrect document cates,whichcouldresultinadelayorreductionin
structureoremphasis. paymentsofyourcertificates."
Hallucinatedcon- Similarsentenceprefixcausesthemodel Gemma-3n-E4B-it,GPT-4o
Documentstartedtorepeatingthesentence"theun-
tinuation tocopythecontinuationofapreviously dergaining and account management standards of
seensentenceinsteadofthecorrectlocal cardissuersandcouldrestricttheabilityofAEN?sof
text. itsaffiliatestotake"onthelastparagraph.
Fixedlexicalcor- Thesamewordortokenisrepeatedly DeepSeek-OCR,
"transferor"waschangedinto"transfessor","trans-
ruption mistranscribedacrosspages. MinerU2.5
foror","transferror",or"transfeder"etc.acrossdiffer-
entdocuments.
Format-sensitive OCRerrorsconsistentlyoccuraround DeepSeek-OCR,
"CertainLegalAspectsoftheReceivables?Consumer
spacing dashes, punctuation, or spacing- MinerU2.5
FinancialProductsRegulation"waschangedinto"Cer-
sensitiveformats. tainLegalAspectsoftheReceivables-ConsumerFi-
nancialProductsRegulation"
Sentence-local AnOCRerrorconfinedtoasinglesen- Llama-family Document missed "and the collateral
securing the
corruption tence,wherethemodelomits,inserts, certificates"
rewrites,ordistortscontentwithinthat
sentencewithoutaffectingotherparts

ofthedocument.
Unstable free- Errorsvarysubstantiallyacrosssamples, Gemma-3n-E4B-it, GPT-
GPT-5missedthewholeparagraphthatevenínclud-
formcorruption even under similar layouts or textual 4.o,GPT-5 ingfinancialfacts.
contexts.
Third,FinCriticalEDevaluatesfactualcorrectnessattheentity
Finally,highlyaccurateOCRmayenablelarge-scalefinancial
level,butdoesnotyetcapturecross-pagelinking,hierarchicalfi-
dataextractionforquestionableuses,includingunauthorizedscrap-
nancialrelationships,orlong-contextnumericalgroundingacross
ing,adversarialmarketstrategies,oramplificationofmisleading
documentcollections. financialnarratives.Researchersandpractitionersshouldconsider
Theselimitationssuggestfutureworkinmultilingual,hybrid
bothbeneficialandharmfuldownstreamimpacts.
human?machineevaluationforedgecases,andricherbenchmarks
G EthicalConsiderationsandLicensing
combiningOCR,layoutunderstanding,andfinancialreasoning.
Alldocumentscomefrompubliclyavailablefinancialfilingsre-
leasedunderopen-accessorresearchlicenses.Noproprietaryor
confidentialinformationisincluded.Thedatasetisintendedfor
F PotentialRisksandMisuse researchondocumentunderstandingandfactualaccuracy;com-
mercialuserequiresseparatecompliancereview.Thebenchmark
FinCriticalED is intended to advance high-precision financial
complieswithACMdataethicspolicies.
OCRresearch,butseveralrisksshouldbenoted.
First,althoughthebenchmarkusespubliclyavailablefinancial Received-;revised-;accepted-
documents,strongerOCRmayenablemoreeffectiveextraction
ofsensitiveinformationfromdocumentsnotintendedforauto-
matedanalysis.Responsibleuserequirescompliancewithprivacy,
regulatory,anddata-handlingrequirements.
Second,theLLM-as-Judgeframeworkisdesignedonlyforevalu-
ationandshouldnotbetreatedasanauthoritativetoolforauditing,
compliance,orlegallybindingdocumentverification.
Third,likeanydataset,FinCriticalEDmayreflectgeographic,
regulatory,andformattingbiasesinitssourcedocuments.Models
trainedortunedonlyonthisdatasetmayoverfittotheseconven-
tionsandgeneralizepoorlyelsewhere.

22. assistant (2026-05-31T04:20:30.794Z)

I now have comprehensive, verbatim data from the paper. I have everything needed to extract falsifiable claims. Let me note the key findings relevant to muneem's research question:

1. **The benchmark itself** ? fact-level OCR on financial documents, 13 systems, key insight that surface OCR metrics mask fact-level errors.
2. **Numbers/monetary units are the most error-prone** ? directly relevant to bank statement amounts/balances.
3. **Specific model rankings** ? Claude-Sonnet-4.6 (cloud) best at 97.23% FFA; among locally-runnable: Qwen3-VL-8B-Instruct 96.88%, GLM-OCR 96.92%, MinerU2.5, etc. GPT-4o surprisingly weak (71.68% FFA, 52.9% critical page error).
4. **Hallucination is documented and characterized** ? hallucinated continuation, transposed/wrong digits (21,829,000 vs 21,429,000), digit/decimal confusion (16,950 vs 16.950).
5. **Verification approach** ? Deterministic-Rule-Guided LLM-as-Judge with deterministic normalization override, which validates muneem's machine-checking instinct.
6. **License note** ? dataset under "open-access or research licenses... commercial use requires separate compliance review" (relevant license caveat).

The strongest local model in the benchmark ? Qwen3-VL-8B-Instruct at 8B / ~96.88% FFA, fits a consumer GPU ? is directly actionable for muneem. Note this paper covers GENERAL financial documents (SEC-style filings), not specifically Indian bank statements, which is a scope caveat I'll flag.

Source quality: this is a primary arXiv research preprint (v3) from a multi-institution academic team (Columbia, Stevens, NYU, Manchester, Harvard, MBZUAI, etc.). Primary.

Let me submit the structured output.

23. user (2026-05-31T04:20:53.216Z)

Structured output provided successfully

24. assistant (2026-05-31T04:21:05.820Z)

I have completed the source extraction. Summary of what I delivered:

Source: FinCriticalED (arXiv:2511.14998), a primary academic preprint from a multi-institution team (Columbia, Stevens Institute, NYU, University of Manchester, Harvard, MBZUAI, et al.). I extracted the full results tables and narrative from the saved PDF using the project venv's pdfplumber (the WebFetch summary alone could not recover the numbers).

Source quality: primary (peer-reviewed-style research preprint with reproducible benchmark, 13 systems, expert-annotated dataset of 859 pages / 9,481 facts).

Five falsifiable claims extracted, all bearing on muneem's research question:

1. **(central)** Numeric values and monetary units are the most error-prone fields on financial docs ? directly relevant to statement amounts/balances. Even a 95.71% ROUGE model gets only 54.05% of monetary units right; GPT-4o hits just 59.56% on numbers.
2. **(central)** Locally-runnable models match cloud: Qwen3-VL-8B-Instruct (8B, fits a consumer GPU) = 96.88% fact accuracy and GLM-OCR (0.9B) = 96.92%, vs. best-overall Claude-Sonnet-4.6 (cloud) at 97.23%. Supports local-first prototyping.
3. **(central)** Documented digit transposition and hallucination (e.g. 21,829,000 vs 21,429,000; 16,950 vs 16.950) ? confirms VLM output must be machine-verified, not trusted.
4. **(supporting)** Frontier cloud VLMs are not uniformly reliable: GPT-4o critical-error rate 52.9% of pages, GPT-5 58.0% ? worse than several local models.
5. **(supporting)** The paper's verification recipe (deterministic normalization + exact-match override, 95% human agreement) directly validates muneem's balance-reconciliation / deterministic-check design.

Key caveat flagged in the output: the benchmark uses SEC-style public financial filings, not Indian bank statements specifically, and the dataset is under "open-access or research licenses" where "commercial use requires separate compliance review" ? a license note worth surfacing given muneem's commercial-safety constraint (this is the benchmark dataset's license, not a model weight license, but relevant context).